

What we're doing

Bulk-port of the protein-docs modeling work that originated on the `marin/protein-training-1b` branch into MarInFold. The entire family (size sweep from 30M to 3B, distance-masked vs unmasked ablation, all-doc-types mixture, continuation script, and matching HF export scripts) is collected here as MarInFold's starting state. Each `train\_protein\_<size>\_<variant>.py` is one ExecutorStep graph, dispatched to iris via MarIn on TPU v5p.

Why

Reproduce the prior research direction on the new infrastructure: does training a small Llama on the `contacts-and-distances-v1-5x` corpus produce a useful protein-structure LM? And what's the best loss formulation + model scale?

Working hypothesis: distance-bin-only loss masking concentrates gradient signal on the prediction of interest and beats unmasked training at matched compute. Scaling should follow Pythia-like curves at least up to 1.4B; beyond that under-training (50K steps × batch 128) is the dominant constraint.

Success criteria

Per size: training reaches the configured step count without divergence; HF export produces a loadable checkpoint at the expected path.

For the sweep: loss vs. params follows a Pythia-like curve on ``eval/protein-docs-cd-val/loss``.

For the loss-mask ablation: distance-masked beats unmasked on ``eval/protein_dist/macro_loss`` at matched checkpoint steps.

Results so far

To be filled in as runs complete on the new Marifold infrastructure. Pre-Marifold results live in W&B under the `marin` project (group `protein-training`); see README.md for the full table of training scripts and baseline run names.

Conclusion

Pending — this is the start state, not a result.